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Question **1**

Not answered

Marked out of 7.00

Theme- Functional Brain Networks

For each of the following descriptions, choose the brain regions from the below list that form a part of the specified functional brain network.

1. Default mode network (Choose THREE)

2. Salience Network (Choose TWO)

3. Central executive network (Choose TWO)

Anterior Insula
Posterior Insula
Superior temporal gyrus
Amygdala
Hippocampus
Medial prefrontal cortex
Posterior cingulate cortex
Precuneus
Angular Gyrus
Dorsolateral prefrontal cortex
Intraparietal sulcus
Occipital Gyrus
Fusiform gyrus
Dorsal anterior cingulate cortex

Explanation: The default mode network includes the posterior cingulate cortex (PCC), precuneus, medial prefrontal cortex (mPFC) and angular gyrus. The dorsal attention network, also called the frontoparietal or central executive network, comprises of dorsolateral prefrontal cortex, Intraparietal sulcus (IPS) as well as lateral superior parietal lobule and the frontal eye field (FEF). The salience network comprises of the anterior prefrontal cortex (aPFC), dorsal anterior cingulate cortex (dACC), anterior insula and frontal operculum, with some subcortical structures (thalamus and putamen).

Correct Answer is :

1. Default mode network - Medial prefrontal cortex, Posterior cingulate cortex, Precuneus

2. Salience Network - Anterior Insula, Dorsal anterior cingulate cortex

3. Central executive network - Dorsolateral prefrontal cortex, Intraparietal sulcus

Question **2**

Not answered

Marked out of 3.00

Theme- Perturb and measure techniques

For each of the research descriptions below choose the most appropriate technique of enquiry from the list below.

1. How does an ion channel react to pharmacological manipulation of the neuron?

DREADD
Optogenetic approach
CLARITY
Patch clamp technique
Extracellular recording
Intracellular recording
Morpholinos
Gene knockout
RNA interference

2. How does a neurotransmitter affect the action potentials of a single neuron?

3. How is neuronal spiking activity coordinated?

Explanation: According to Shieh and Carter, the three main categories of electrophysiology techniques are defined by where the recording instrument, the electrode, is placed in the neural specimen. "In an extracellular recording experiment, the electrode is placed just outside the neuron of interest. This is best suited to measure neuronal spiking and summed postsynaptic potentials of a set of cells in the vicinity (local field potentials). In an intracellular recording experiment, the electrode is inserted into the neuron of interest. This can be used to measure subthreshold changes in potentials such as EPSPs and IPSPs. Finally, using patch clamp techniques, the electrode is placed close to the neuronal membrane, forming a tight seal with a patch of the membrane. These different recording techniques are used to examine the electrical properties of neurons both in vitro and in vivo. Questions regarding signals from neurons in vivo are most easily addressed using extracellular methods. Questions regarding the open and closed states of a specific ion channel in the presence of neurochemical agents are best addressed using patch clamp techniques". Ref: Excerpt from Carter & Shieh, Guide to Research Techniques in Neuroscience. Academic Press, 2010

Correct Answer is :

1. How does an ion channel react to pharmacological manipulation of the neuron? - Patch clamp technique

2. How does a neurotransmitter affect the action potentials of a single neuron? - Intracellular recording

3. How is neuronal spiking activity coordinated? - Extracellular recording

Question **3**

Not answered

Marked out of 7.00

Theme- Genomic engineering techniques

Which of the listed techniques are associated with the below descriptions?

1. Nuclease based gene targeting techniques (Choose THREE)

2. Silencing transcribed gene products (Choose ONE)

3. Preventing ribosomal access for protein translation (Choose ONE)

Conditional Knockouts
CRISPR
TALEN
Targeted Gene Knockouts
DREADD
CLARITY
Morpholinos
ZFN
Gene knockout
RNA interference

4. Replacing a functional gene with non-functional sequence in the embryonic cells (Choose ONE)

5. Cell-type specific replacement of functional genes (Choose ONE)

When a functional gene is knocked out in the entire organism, this could be lethal at the embryonic stage. Alternatively, the knockout can be tissue or cell-type specific or done in adult animals to prevent lethal effects. In such **conditional knockouts**, a gene is removed from the genome in a precise spatial or temporal manner.

Animals with tissue specific promoters (Cre) are bred with animals that carry the functional gene flanked by flagging sequences (floxed sequence). Modern gene editing tools such as **CRISPR, Zinc finger nucleases (ZFNs), and Transcription activator-like effector nucleases (TALENs)** create a double-strand break at a precise location in the genome using an enzyme that cleaves nucleic acids (nuclease). The nuclease is directed to a specific target in the genome by interacting with a guide strand of DNA or RNA.

Instead of knocking out a gene at the DNA level, we can also silence the expression of the gene at the RNA level. This is called **RNA interference** (RNAi). RNAi is an endogenous process used to regulate gene expression and to degrade foreign mRNA from phages. A double-stranded RNA (dsRNA) is cleaved by an enzyme called dicer resulting in small duplex fragments, (~ 19-23 base pairs long), called small interfering RNA (siRNA). The siRNA-Dicer forms a RNA-induced silencing complex (RISC) which further degrades RNA strands that have a complementary nucleotide sequence to one of the siRNA guide strands. We can use a dsRNA molecule homologous to the mRNA transcript to be silenced to achieve the intended RNAi. We can also deliver synthetic siRNAs to attract the RISC complex and bind the homologous mRNA transcript. Also, short hairpin RNA (shRNA) with complementary sequences at either end, can cause the molecule to fold onto itself, producing a hairpin shape.

Morpholinos are synthetic 22-25 bp antisense oligonucleotide analogues designed to complement an RNA sequence. A morpholino and its complementary mRNA will hybridize to each other, preventing a ribosome from accessing the mRNA and translating it into a protein. Unlike RNAi, morpholinos do not degrade their target mRNA molecules.

Genetic knockout mice are created in the laboratory by injecting the genetic construct prepared using DNA recombinant technology into embryonic stem (ES) cells and inserting successfully targeted ES cells into a blastocyst, placed inside a female mouse for breeding the offspring. This is a long process, that can take up to 2 years before we can create a genetic knockout breed.

Ref: Excerpt from Carter & Shieh, Guide to Research Techniques in Neuroscience. Academic Press, 2010

Correct Answer is :

1. Nuclease based gene targeting techniques - CRISPR, TALEN, ZFN

2. Silencing transcribed gene products - RNA interference

3. Preventing ribosomal access for protein translation - Morpholinos

4. Replacing a functional gene with non-functional sequence in the embryonic cells - Targeted Gene Knockouts

5.Cell-type specific replacement of functional genes - Conditional Knockouts

Question **4**

Not answered

Marked out of 1.00

Flag question

CRISPR is a technique that is used to make specific changes in the DNA of plants and animals. CRISPR stands for

Select one:

- Cutting Recurring Introns and Selectively Pasting Repeats
- Cutting Regularly Interspaced Small Purposeless Repeats
- Clustered Regularly Interspaced Short Protein Rings
- Cleaving Randomly Interleaved Silencing Protein Repeats
- Clustered Regularly Interspaced Short Palindromic Repeats

CRISPR is the commonly known acronym for the genome editing technique Clustered Regularly Interspaced Short Palindromic Repeats. Ref: Heidenreich M, Zhang F. Applications of CRISPR–Cas systems in neuroscience. Nature Reviews Neuroscience. 2016 Jan;17(1):36.

The correct answer is: Clustered Regularly Interspaced Short Palindromic Repeats

Question **5**

Not answered

Marked out of 1.00

Flag question

Which of the following is a major challenge for the use of CRISPR technique to alter neuronal gene expression in human adult brains?

Select one:

- Ischemic response to the use of Cas gene in humans
- Blood-Brain Barrier
- Autoimmune inflammatory reaction to Cas
- Epigenetic effects on the human genome
- Lack of knowledge regarding human genome sequences

The major obstacle for employing CRISPR as a gene therapy in neuroscience is our inability to safely overcome the blood brain barrier (BBB) and reach the specifc cells that are affected by a genetic aberration. Approaches such as the use of adenovirus vehicles have fallen short to date as the Cas gene and the other biological components required for gene editing through CRISPR constitute a much larger baggage than what adenoviruses can effectively carry past the BBB. Ref: SenÍs E, Fatouros C, Große S, Wiedtke E, Niopek D, Mueller AK, Börner K, Grimm D. CRISPR/Cas9-mediated genome engineering: an adeno-associated viral (AAV) vector toolbox. Biotechnology journal. 2014 Nov;9(11):1402-12.

The correct answer is: Blood-Brain Barrier

Question **6**

CRISPR-associated protein system Cas is best described as

Not answered

Marked out of 1.00

Flag question

Select one:

- an innate sequence defending against epigenetic changes
- a viral vector delivery system
- an innate microbial defense mechanism against pathogens
- DNA-guided RNA editing sequence
- a RNA dependent RNA polymerase

The major function of CRISPR-associated (cas) genes for the bacteria that carry them is to provide innate immunity against the invasion of foreign nuclei acids such as plasmids and phages (viruses). Upon the first exposure to such foreign agents, short fragments of the foreign DNA (called protospacers) get incorporated into the CRISPR gene's repeat-spacer array (short palindromic repeats) as new spacers (usually of 20 nucleotides length) . This becomes the genetic memory of the prior invasion. Transcription of this array results in short mature CRISPR RNAs (crRNAs) that carry the tag (the RNA complement of the protospacer). Upon second exposure, the crRNA spacer 'sticks' to the target sequence of the foreign body and triggers sequence-specific destruction of theinvader using endonucleases. There are several types (1 to 5) of CRISPR systems identified to date. OF these, the type II CRISPR systems (Cas9) has a seed region called protospacer-adjacent-motif (PAM) that consists of 2 to 5 basepairs at the 3' end of the 20-nucleotide long spacer sequence. PAM is critical for the target selection and degradation. Ref: Heidenreich M, Zhang F. Applications of CRISPR–Cas systems in neuroscience. Nature Reviews Neuroscience. 2016 Jan;17(1):36.

The correct answer is: an innate microbial defense mechanism against pathogens

Question **7**

Not answered

Marked out of 1.00

Flag question

The most commonly used system for CRISPR based genome editing is

Select one:

- System I
- System II
- System III
- System V
- System IV

There are several types (1 to 5) of CRISPR systems identified to date. Of these, the type II CRISPR systems (Cas9) has a seed region called protospacer-adjacent-motif (PAM) that consists of 2 to 5 basepairs at the 3' end of the 20-nucleotide long spacer sequence. PAM is critical for the target selection and degradation. Ref: Heidenreich M, Zhang F. Applications of CRISPR–Cas systems in neuroscience. Nature Reviews Neuroscience. 2016 Jan;17(1):36.

The correct answer is: System II

Question **8**

Not answered

Marked out of

Which of the following elements of CRISPR gene editing produces hindrance to DNA expression without causing a degradation?

1.00

Flag question

Select one:

- spacer
- dead Cas9
- protospacer
- PAM
- short Cas9

While the Cas9 gene used in routine CRIPSR experiments is a nuclease that cleaves DNA, CRISPR can also be used for gene silencing (i.e. reversible repression), gene activation (inducing expression). When the 2 catalytic domains of Cas9 are inactivated, this produces dead Cas9 (dCas9). dCas9 does not cleave DNA but produces stereological hindrance, resulting in a phenomenon known as CRISPR interference (CRISPRi).

The correct answer is: dead Cas9

Question **9**

Not answered

Marked out of 1.00

Flag question

You are reviewing a graduate student who is seeking help for the treatment of his longstanding anxiety. He has recently come across procedures that allow brains to be "grown in a dish" through a novel stem-cell technology. Which of the following refers to cells that are reprogrammed to an embryonic stem cell-like state?

Select one:

- hiPSC
- zinc-finger cells
- PAM
- trans-activating CRISPR
- Cas9

Well-differentiated adult somatic cells such as fibroblasts from the human skin can be obtained from a donor, and reprogrammed to become pluripotent cells. These human induced pluripotent stem cells (hiPSC) have properties similar to those of embryonic stem cells. They can be further differentiated into any other cell type. Furthermore, gene editing can be performed on these hiPSC, providing a means to observe the effect of genetic mutations across various cellular lineages. While cells such as neurons are difficult to grow in a dish, hiPSCs have enabled the culture of neurons much more accessible.

The correct answer is: hiPSC

Question **10**

Not answered

Marked out of 1.00

Flag question

Which of the following refers to a mechanism that is highly conserved across species and serves to repair damage to DNA?

Select one:

- Homology-directed repair
- double-strand break (DSB) mechanism
- transcription activator-like effectors

- Zinc-Finger nucleases
- trans-activating CRISPR

When nucleases such as zinc-finger nuclease, TALENs or Cas result in DNA damage, one of the 2 highly conserved repair mechanisms gets triggered. Non-homologous end-joining (NHEJ) and homology-directed repair (HDR) are the 2 approaches that follow DNA double-strand breaks (DSB) induced by nucleases. These repair processes ensure prevention of cell death following DNA damage. Of these, NHEJ is very much prone to errors, and often results in various lengths of insertions and deletions (called "indels") and thus frameshift mutations that lead to gene knockout. On the other hand, HDR utilizes a homologous donor DNA template that accurately corrects the DSB. Gene editing techniques exploit the mechanism of HDR to introduce specific sequences into the genome, thus engineering the required genetic mutations. ZFNs and TALENs use protein domains to enable DNA binding, thus requiring tailored synthesis of nucleases for each target site. Cas9 is guided by the tag carried by CRISPR RNA (crRNA)) that is associated with the effector complex (trans-activating CRISPR RNA nuclease) crRNA (tracrRNA), making HDR readily accessible. A Cas9 system derived from Streptococcus pyogenes or Staphylococcus aureus can be used to cleave any genomic locus of interest.

The correct answer is: Homology-directed repair

Question **11**

Not answered

Marked out of 1.00

Flag question

In gene editing approaches, the term gRNA refers to

Select one:

- gnome RNAs
- generic RNAs
- global RNA sequences
- guide RNAs
- gliding RNAs

RNAs that guide the endonucleases that cleave DNA to form double strand breaks are called guide RNAs or simply gRNAs.

The correct answer is: guide RNAs

Question **12**

Not answered

Marked out of 1.00

Flag question

Which of the following is an advantage of CRISPR technology compared to other gene editing tools?

Select one:

- can be used to alter pluripotent cells
- multiple genetic targets can be edited
- easy access via blood-brain barrier
- readily transmitted using adenoviral vectors to adult brains
- larger chunks of DNA can be traced

Compared to older gene editing technologies, such as zinc-finger nucleases (ZFNs) and transcription activator-like effector nucleases (TALENs), CRISPR offers several advantages. It provides a faster means to edit multiple genetic targets in one experiment. In addition, it is cheaper as well. CRISPR technique involves RNA guided DNA sequence detection, while ZNF and TALENs involve protein guided detection. As a result, ZFN and TALENs need to be tailored for each search, while CRISPR is more accessible. ZFN and TALENs can also be used to edit hiPSC, though CRISPR provides easier means to achieve this. Ref: Cong L, Ran FA, Cox D, Lin S, Barretto R, Habib N et al (2013). Multiplex genome engineering using CRISPR/Cas systems. Science 339: 819–823.

The correct answer is: multiple genetic targets can be edited

Question **13**

Not answered

Marked out of 1.00

Flag question

Endonucleases used for gene editing result in

Select one:

- double-strand break (DSB)
- activation of growth factor transcription
- translocation of genetic material
- cell death
- epigenetic changes with knockout effect

Site-specific nucleases, including ZFNs, TALENs and Cas proteins, enable precise genetic modifications by inducing double-strand DNA breaks (DSBs) at target locations in the genome. Ref: Heidenreich M, Zhang F. Applications of CRISPR–Cas systems in neuroscience. Nature Reviews Neuroscience. 2016 Jan;17(1):36.

The correct answer is: double-strand break (DSB)

Question **14**

Not answered

Marked out of 1.00

Flag question

With respect to the electromagnetic fields recorded from the scalp surface, which of the following is the most prominent source of the recorded activity?

Select one:

- electrical flow tangential to pyramidal neurons
- asynchronous burst of the closest single neuronal cell
- electrical activation of interneurons
- post-synaptic electrical flow along the pyramidal dendrites
- pre-synaptic currents on pyramidal soma

Radial currents produce magnetic fields that are contained within the brain tissue (right hand rule of electromagnetism). As a result, MEG only detects tangential currents, produced by tangential currents, parallel to scalp surface. But perfectly radial neurons that are exactly perpendicular to scalp surface are rather few. As a result, MEG can generate representative maps of neural activity.

The correct answer is: post-synaptic electrical flow along the pyramidal dendrites

Question **15**

Not answered

Marked out of 1.00

Flag question

Which of the following neuroimaging techniques require a helium filled tank to enhance superconductivity?

Select one:

- EEG
- NIRS
- MEG
- MRS
- PET

Superconducting QUantum Interference Device or SQUID allows the detection of the brain's weak magnetic fields without the need for an electric reference. To enable superconductive properties, the SQUIDs have to be kept at -269 °C. Liquid helium filled tanks are required for this purpose. Modern MEG machines contain helmet-shaped arrays of more than 300 SQUID sensors immersed in liquid helium. Ref: Hari R, Salmelin R. Magnetoencephalography: from SQUIDs to neuroscience: neuroimage 20th anniversary special edition. Neuroimage. 2012 Jun 1;61(2):386-96.

The correct answer is: MEG

Question **16**

Not answered

Marked out of 1.00

Flag question

Which of the following is true regarding MEG?

Select one:

- EEG signal is dominated by currents flowing tangential to the skull.
- MEG signal comprises of the fields induced by both radial and tangential currents
- MEG signal is dominated by currents flowing tangential to the skull.
- EEG signal comes exclusively from currents flowing perpendicular to the skull.
- MEG signal is dominated by currents flowing perpendicular to the skull.

Radial currents produce magnetic fields that are contained within the brain tissue (right hand rule of electromagnetism). As a result, MEG only detects tangential currents. But perfectly radial neurons that are perpendicular to scalp surface are rather few. As a result, MEG can generate representative maps of neural activity.

The correct answer is: MEG signal is dominated by currents flowing tangential to the skull.

Question **17**

Not answered

Marked out of 1.00

A group of researchers derived hiPSCs from the fibroblasts of patients with Rett syndrome. They found that the patient-derived neurons had fewer synapses, reduced spine density, and smaller cell body compared to wild type controls. Which of the following mutations are likely to be present in the hiPSC lineages?

Flag question

Select one:

- DISC1 mutations
- CDKL5 mutations
- MECP2 mutations
- FOXG1 mutations
- CDH8 mutatoins

Rett's syndrome exclusively affects females (as male mutant foeuses do not survive to full gestation). Rett syndrome is most often caused by X-linked mutations in the MECP2 gene. MECP2 is a transcription factorthat plays a key role in the normal brain development. Rett syndrome is also associated with mutations in CDKL5 or FOXG1 though the latter are less common.

The correct answer is: MECP2 mutations

Question **18**

Not answered

Marked out of 1.00

Flag question

Patch-clamp recording is best used to measure

Select one:

- Error related negativity
- Mismatch negativity
- Single channel currents
- Local field potential
- Evoked response potentials

The patch-clamp technique is an electrophysiological approach to measure the currents passing through ion channels in the cell membrane. Developed by Neher and Sakmann in 1976, based on the experiments of odgkin and Huxley in the 1950s, patch clamp technique allows us to measure ion-channel activities even at the level of a single channel.

The correct answer is: Single channel currents

Question **19**

Not answered

Marked out of 1.00

Flag question

Which of the following neuroscietific approach can be used to activate or inhibit specific neuronal projections in live animals using light exposure?

Select one:

- Photostimulation
- Flurostimulation
- Optogenetics
- Deep brain stimulation
- DREADD

Genetically encoded light sensitive protein channels are used in optogenetics. Depending on the specific pigment protein that is engineered genetically to be expressed in an organism (rodents, primates), depolarisation or hypopolarisation of specific neuronal populations can be achieved using specific frequency of light. Light is usually delivered very close to the target neurons due to the scattering effect of brain tissue, and tis is achieved usually by implanting a lightweight fiber optic cable guided by a surgical cannula.

The correct answer is: Optogenetics

Question **20**

Not answered

Marked out of 1.00

Flag question

When employing optogenetics to study neuronal functions, which of the following light will activate the pigment rhodopsin?

Select one:

- Blue light
- Red light
- Green light
- White light
- Yellow light

Channelrhodopsin-2 (ChR2) is a cation channel that opens in response to blue light. Halorhodopsin, from the halobacterium Natronomonas pharaonis, is a chloride channel that actively moves chloride ions into the cell from extracellular zone when yellow light is presented. Archaerhodopsin, derived from Halorubrum sodomense, is a proton pump that actively removes hydrogen ions from the cell when a yellow light is presented. There are two major transgenes used for optogenetic inhibition. Both halorhodopsin and archaerhodopsin serve to hyperpolarize neurons, while channelrhodopsin promotes depolarization.

The correct answer is: Blue light

Question **21**

Not answered

Marked out of 1.00

Flag question

The most common adverse effects of rTMS is

Select one:

- Manic switching
- Vertigo
- Seizures
- Headaches
- Syncope

The most common adverse effects of rTMS includes scalp pain during stimulation (35%–40%), and transient headache after stimulation (25%–30%). It is important to note that the frequency of these side effects vary with the stimulation protocol. In addition, these symptoms are often transient, diminish during the couse of treatment and respond well to over-the-counter analgesics. The all-causes discontinuation rate for rTMS is estimated to be about 5%. Ref: http://www.cmaj.ca/content/188/16/1175

The correct answer is: Headaches

Question **22**

Not answered

Marked out of 1.00

Flag question

A neuroscientist working in your academic unit describes how she places a micropipette against a cell and apply suction to form a high resistance seal. She then uses an electrode within the micropipette to control the electrical properties of the cell membrane and studies ion channel activity. This technique is called

Select one:

- Patch clamp recording
- Diffusion weighted recording
- Intracellular recording
- Confocal recording
- Extracellular recording

The patch-clamp technique helps us to study single or multiple ion channel functions in live tissue. Whole-cell patch-clamp recordings allow the study of the electrical behavior of a substantial part of a neuron in vitro on brain slices, or freshly dissociated neurons, or on cell culture models.

The correct answer is: Patch clamp recording

Question **23**

Not answered

Marked out of 1.00

Flag question

CLARITY is the tem used to describe which of the following neuroscientific methods?

Select one:

- Diffusion tensor imaging
- Magnetic resonance spectroscopy
- Hydrogel tissue clarification
- Artifical intelligence
- Embryonic stem cell culture

Imaging of whole brains with cellular resolution is crucial to study neuronal morphology, connectivity and phenotyping. CLARITY, an approach introduced recently by Karl Deisseroth at Stanford produces structurally intact, yet optically transparent tissue,that can be imaged without sectioning. It involves a process by which the light-scattering lipids in the brain tissue are replaced by a transparent hydrogel matrix. Ref: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4586927/

The correct answer is: Hydrogel tissue clarification

Question **24**

Not answered

Marked out of 1.00

All of the following brain structures constitute the salience network, except

Select one:

- thalamus
-

Flag question

anterior insula

- Precuenus
- dorsal anterior cingulate
- striatum

Precuneus is a part of the medial superior parietal lobule. It is a major brain hub tht constitutes a key node in the default mode, not the salience, network.

The correct answer is: Precuenus

Question **25**

Not answered

Marked out of 1.00

Flag question

Which of the following best described the neuropsychological function of the default mode network?

Select one:

- Introspection
- Reward orientation
- Task execution
- Interoception
- Interference

The DMN is “an organized spontaneous network of neural activity that is modulated during attention-demanding cognition which is characterized by spontaneous BOLD signal fluctuations which tend to inversely correlate with fluctuations in other networks, including those that subserve arousal, attention, perception, and working memory". While the description of DMN is largeley based on what it does not do (task-based engagement), recent studies have implicated DMN to have a role in introspection, mind-wandering, daydreaming as well as autobiographical recall.

The correct answer is: Introspection

Question **26**

Not answered

Marked out of 1.00

Flag question

The term task-negative network refers to

Select one:

- The auditory network
- The default mode network
- The executive network
- The sensorimotor network
- The frontoparietal network

The DMN includes those regions that are deactivated during performance of a task when compared to the resting, baseline condition (Raichle et al., 2001). Greicius et al. (2003) linked resting-state functional connectivity methodology to the default mode hypothesis, and coined

the term “DMN". Greicius et al.‘s (2003) article, which is commonly thought to have launched the DMN, noted that certain brain regions “consistently show greater activity during resting states than during cognitive tasks" [excerpt from https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4142807/]

The correct answer is: The default mode network

Question **27**

Not answered

Marked out of 1.00

Flag question

Which of the following is true regarding the use of transcranial magnetic stmulation to treat the elderly with depression?

Select one:

- The evidence is firmly against its use in the elderly
- The evidence is firmly supportive of its use in the elderly
- Due to cortical atrophy, rTMS cannot deliver effective pulses to the cortex of patients older than 65
- The evidence is favourable only for men not women older than 65
- The evidence is insufficient due to exclusion of older patients in most RCTs

The correct answer is: The evidence is insufficient due to exclusion of older patients in most RCTs

Question **28**

Not answered

Marked out of 1.00

Flag question

Which of the following is true regarding the use of transcranial magnetic stmulation to treat treatment-resistant depression?

Select one:

- It is not superior to ECT in terms of efficacy
- It works as well as ECT for psychotic depression
- It works better than ECT for melancholic depression
- It produces more cognitive dysfunction than ECT
- It causes more side effects that ECT

A meta-analysis of head-to-head trials of electroconvulsive therapy v. rTMS (n = 425 patients, 9 trials) concluded that ECT is superior to rTMS with 64.4% v. 48.7% response and 52.9% v. 33.6% remission rates. In particular, in the presence of psychotic symptoms, ECT was much superior. Nevertheless, in certain domains of side effects (esp. memory disturbances), rTMS was much more tolerable. Current treatment pathways endorse rTMS efficacy to be comparable or higher than that of psychotherapy and medications, but rather lower than ECT, when considering sequential treatment after successive treatment failures for major depressive disorder.

The correct answer is: It is not superior to ECT in terms of efficacy

Question **29**

The brain region targeted by deep brain stimulation for addressing treatment resistant OCD

Not answered

Marked out of 1.00

Flag question

is

Select one:

- anterior limb of the internal capsule
- Amygdala
- Dorsolateral prefrontal cortex
- Hippocampus
- Arcuate fasciculus

In OCD, DBS is often targeted on “striatal areas", including the anterior limb of the internal capsule (ALIC), the ventral capsule and ventral striatum (VC/VS), the nucleus accumbens (NA) or the ventral caudate nucleus, subthalamic nucleus, or the inferior thalamic peduncle.

The correct answer is: anterior limb of the internal capsule

Question **30**

Not answered

Marked out of 1.00

Flag question

When functional connectivity is measured using resting state fMRI, the frequency of BOLD signal fluctuations captured by the technique is

Select one:

- >10Hz
- <0.1Hz
- >1Hz
- >2.5Hz
- <0.01Hz

Resting state fcMRI measures spontaneous, high-amplitude, low-frequency (<0.1 Hz) BOLD signal fluctuations in subjects at rest (i.e., when not involved in any explicit task).

The correct answer is: <0.1Hz

Question **31**

Not answered

Marked out of 1.00

Flag question

Which of the following network is said to play a switching function in flipping the brain from a resting to cognitively active state?

Select one:

- The sensorimotor network
- The frontoparietal network
- The default mode network
- The executive network
- The salience network

The DMN is associated with self-referential mental activity. It is also describe to be activated

by “stimulus-independent thought". When explicit task processing cognitive demands are placed, brain regions constituting the DMN get switched off or appear inactive. The switching function between DMN and task-positive networks such as frontoparietal regions is thought be managed by the salience network.

The correct answer is: The salience network

Question **32**

Not answered

Marked out of 1.00

Flag question

Mr. Robert Gove is a 53 year old patient with OCD who is resistant to conventional pharmacological treatment approaches. He is now receiving an experimental treatment that is providing notable symptom relief. He describes how a pacemaker implanted under his collarbone is "clicked on" by his psychiatrist during regular visits. Which of the following treatments is he receiving?

Select one:

- Transcranial alternate current stimulation
- Subcutaneous electrical non-convulsive stimulation
- Deep brain stimulation
- Vagal nerve stimulation
- Gamma ventral capsulotomy

In DBS, the electrodes placed in the brain are connected by subcutaneous wires to implanatable neurostimulators (pulse generating batteries with a programmable microchip) placed below the collarbone. The treatment provider can use a hand-held wand to operate the battery.

The correct answer is: Deep brain stimulation

Question **33**

Not answered

Marked out of 1.00

Flag question

Which of the following is an advantage of human induced pluripotent stem cells compared to embryonic stem cells?

Select one:

- Allows in vivo experiments
- Genome editing is feasible
- Ethically acceptable
- Only dead embryos are used
- Low possibility of mutations during in vitro culturing

Pluripotent stem cells, when cultured, are capable of repeated self–renewal as well as differentiation to all adult cell types. Pluripotency of these cells results from a specific transcriptional gene network (especially Oct4, Sox2 , and Nanog genes). One of the well known source of pluripotent stem cells is the embryo. Embryonic stem cells (ESCs) are obtained from blastocyst stage of embryos. Research using ESCs has several ethical concerns, especially as retrieval of blastocyst cells leads to embryonal death. Since 2006, it has bcome possible to generate pluripotent cells by programming adult somatic cells, which results in induced pluripotent stem cells (iPSCs). These induced PSCs closely resemble

ESCs with preserved pluripotency. As no embryos are required, this procedure is much more acceptable for ethical reasons.

The correct answer is: Ethically acceptable

Question **34**

Not answered

Marked out of 1.00

Flag question

The term tDCS refers to

Select one:

- Transcranial Dorsolateral Cortical Stimulation
- Transcranial Direct Current Stimulation
- Transcranial Deep Circuit Stimulation
- Treatment with Dynamic Cyclical Stimulation
- Transcranial Delta Charged Stimulation

Transcranial direct current stimulation (tDCS) and transcranial alternating current stimulation (tACS) are stimulation techniques that modulate neural activity in the human brain in a non-invasive manner using weak electrical currents.

The correct answer is: Transcranial Direct Current Stimulation

Question **35**

Not answered

Marked out of 1.00

Flag question

Which of the following sequence of stimulation, when applied to the DLPFC, can reduce the GABA to glutamate concentration ratio?

Select one:

- intermittent beta-burst stimulation
- intermittent gamma-burst stimulation
- continuous beta-burst stimulation
- continuous theta-burst stimulation
- intermittent theta-burst stimulation

Dorsolateral prefrontal intermittent theta-burst stimulation (iTBS), in healthy controls, has been shown to reduce GABA/Glutamate ratio. Ref: Iwabuchi SJ, Raschke F, Auer DP, Liddle PF, Lankappa ST, Palaniyappan L. Targeted transcranial theta-burst stimulation alters fronto-insular network and prefrontal GABA. Neuroimage. 2017 Feb 1;146:395-403.

The correct answer is: intermittent theta-burst stimulation

Question **36**

Not answered

Marked out of 1.00

Flag question

The term gyrification refers to the developmental process of

Select one:

- Cortical folding
- Cerebellar development

- Subcortical connectivity
- Neuronal migration
- Sulcal widening

Gyrification is a developmental process by which the folding of cortical surface occurs leading to distinct patterns on the brain. Gyrification results in a significant increase in the cortical surface area during foetal development.

The correct answer is: Cortical folding

Question **37**

Not answered

Marked out of 1.00

Flag question

Which of the following is more or less completed by the age of 2 in normally developing children?

Select one:

- Cortical contraction
- Synaptic pruning
- Cortical thinning
- Cortical gyrification
- Cortical myelination

The development of gyrification begins prior to birth, with the early stages of gyral and sulcal formation taking place between 10 to 15 weeks of human fetal life (Zilles et al., 1997). During the third trimester of fetal life, when the brain is undergoing considerable growth (Chi et al., 1977), the brain develops from a relatively smooth, lissencephalic structure to a brain that more closely resembles the morphology of the adult brain. Most gyrification is completed in utero, with late neonatal changes completed by the age of 2. Using histological approaches, it has been demonstrated that the time-course for the laydown of synapses (synaptic blooming) and their elimination (synaptic pruning) in the human brain varies enormously by brain region. In the visual cortex (and many primary sensory areas), synaptic overproduction is maximal at about the fourth postnatal month, followed by synapse elimination. But in the medial prefrontal cortex (and many association areas), the peak occurs later at 3–4 years of age, with most decline occurring in the mid-to-late adolescence. Cortical thickness shows a linear decresae from the age of around 4 onwards in a steady monotonic manner, with an acceleration during later ages (>60) (Ducharme et al., 2014, Sowell et al., 2004, Muftuler et al., 2011, Tamnes et al., 2010, Koolschijn and Crone, 2013). Cortical thickness has later peaks around the ages 8-9, in association areas (posterior parietal areas, posterior temporo-parietal areas, and right DLPFC (Shaw et al., 2008, Mutlu et al., 2013)).. At present, the cellular drivers of thickness changes are not known. It is suspected that both synaptic pruning and progressive intracortical myelinatio with age could play a role. Myelination continues well into the third decade of life, with a ventral-to-dorsal pattern, with the most dorsal regions myelinating continually during adolescence. Subcortical structures (especially the caudate) also undergo progressive volume loss in the deep gray-matter nuclei around puberty. In contrast to the basal ganglia, medial temporal lobe gray-matter structures (such as the amygdala and hippocampus) appear to increase in volume during childhood and adolescence. Ref: Ducharme, Simon, et al. "Trajectories of cortical thickness maturation in normal brain development—the importance of quality control procedures." Neuroimage 125 (2016): 267-279. Paus T. Mapping brain maturation and cognitive development during adolescence. Trends in cognitive sciences. 2005 Feb 1;9(2):60-8.

https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3113697/ and
https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2815169/

The correct answer is: Cortical gyrification

Question **38**

Not answered

Marked out of 1.00

Flag question

With respect to functional connectivity, which of the following statements is correct?

Select one:

- It is static and does not change with time
- It is fixed and does not change during development
- It can only be quantified using functional MRI
- It refers to correlation in the time course of activation between 2 brain regions
- Structural connectivity is essential to establish functional connectivity

The term “functional connectivity" refers to the statistical dependency (usually, a correlation) between 2 or more spatially-distinct brain regions. Functional connectivity is often interpreted to represent simultaneous engagement among the brain regions when performing in a task. Functional connectivity may exist even if no strucutral (white matter based) connections can be traced between 2 brain regions. It can be quantified using EEG, PET, fMRI or MEG, though fMRI is the most commonly used tool. Functional connectivity is dynamic, i.e. 2 regions may show functional connectivity during a specific time window but not connected at other times. During development, a redistribution of functional ocnnectivity takes place, with anatomically distant sites getting connected at a functional level.

The correct answer is: It refers to correlation in the time course of activation between 2 brain regions

Question **39**

Not answered

Marked out of 1.00

Flag question

Which of the following brain networks is most likely to have regions that are deactivated when performaing a cognitively demanding task?

Select one:

- The default mode network
- The sensorimotor network
- The executive network
- The frontoparietal network
- The salience network

The DMN is associated with self-referential mental activity. It is also describe to be activated by “stimulus-independent thought". When explicit task processing cognitive demands are placed, brain regions constituting the DMN get switched off or appear inactive. The switching function between DMN and task-positive networks such as frontoparietal regions is thought be managed by the salience network.

The correct answer is: The default mode network

Question **40**

Not answered

Marked out of 1.00

Flag question

The frequency of transcranial magnetic stimulation that is shown to be effective and most widely used in treating depression is

Select one:

- 60Hz
- 40Hz
- 10 Hz
- 2Hz
- 1Hz

US FDA approved a protocol of left DLPFC stimulation at 10 Hz frequency (10 pulses/sec, 120% of motor threshold, 3000 pulses/session, 5 times a week for 4-6 weeks). This is the most widely used protocol to date for treating depression, but increasingly novel protocols such as theta-burst stimulation require just one to three minutes, and have shown favourable results in early studies.

The correct answer is: 10 Hz

Question **41**

Not answered

Marked out of 1.00

Flag question

The brain region targeted by transcranial magnetic stimulation for addressing treatment resistant depression is

Select one:

- Anterior cingulate cortex
- Amygdala
- Dorsolateral prefrontal cortex
- Habenula
- Motor cortex

US FDA approved a protocol of left DLPFC stimulation at 10 Hz frequency (10 pulses/sec, 120% of motor threshold, 3000 pulses/session, 5 times a week for 4-6 weeks). This is the most widely stimulated target site to date for treating depression, but increasingly other regions such as dorsomedial prefrontal cortex and insula (via deep TMS) have also been targeted with some encouraging results in early studies.

The correct answer is: Dorsolateral prefrontal cortex

Question **42**

Not answered

Marked out of 1.00

Flag question

The brain region targeted by deep brain stimulation for addressing treatment resistant depression is

Select one:

- Amygdala
- Hippocampus
- Anterior cingulate cortex

- Habenula
- Dorsolateral prefrontal cortex

To date, studies of DBS in depression have primarily targeted the subcallosal cingulate gyrus (Brodmann Area 25 or Cg25 or Meyberg site), ventral capsule/ventral striatum, medial forebrain bundle, and nucleus accumbens. The best reported effect size to date has been obtained from stimulating the subcallosal cingulate cortex (Cg25). The ventral capsule/ventral striatum (VC/VS) site was accidentally discovered as a potential site when conducting DBS trials on OCD, as mood improvement preceded change in OCD symptoms when VC/VS was stimulated.

The correct answer is: Anterior cingulate cortex

Question **43**

Not answered

Marked out of 1.00

Flag question

Estimating a subject specific motor threshold before prescribing a treatment dose is required for which of the following techniques?

Select one:

- tDCS
- MEG
- TMS
- ECT
- tACS

Transcranial magnetic stimulation utilizes the measurement of motor threshold. Motor Threshold is defined as the minimum TMS intensity sufficient to produce a predefined motor-evoked potential (MEP) in the contralateral abductor pollicis brevis in at least 50% of trials [Rossini et al., 1994]. MT, roughly a measure of the TMS intensity necessary to evoke a peripheral motor response, is highly variable across individuals but also remarkably constant in a given individual [Cicinelli et al., 1997; Mills and Nithi, 1997; Ziemann et al., 1996]. The MT is of particular importance in TMS experimental design as a way of calibrating and normalizing TMS coil output energy for individual physiologic variability and thus determines both dose and safety limits. Excerpt from https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2893589/

The correct answer is: TMS

Question **44**

Not answered

Marked out of 1.00

Flag question

Clozapine-N-oxide, a pharmacologically inert metabolite of the atypical antipsychotic drug clozapine, is a Designer Drug (DREADD) used for enhancing neuronal signalling. Which of the following receptors does it predominantly bind to in experimental studies?

Select one:

- M1 Muscarinic
- D4 Dopaminergic
- M2 Muscarini
- D2 Dopaminergic

M3 Muscarinic

Designer Receptors Exclusively Activated by Designer Drugs (DREADD) is a chemogenetic tool that allows targetted manipulation of neuronal cells through genetically engineered expression of designer receptors that can be activated only by specific designer drugs. Clozapine-N-oxide (CNO) is a designer drug that is inert pharmacologically but activate a specific designer G-protein coupled (Gq-DREADDs) receptors. In particular, CNO acts on the muscarininc M3 designer receptor. It can also activate designer M1 and M5 receptors although hM3Dq is the most frequently used. While inert at the recommended doses (0.1–3 mg/kg), CNO may get back-transformed to clozapine via metabolic pathways but this occurs at a low level.

The correct answer is: M3 Muscarinic

Question **45**

Not answered

Marked out of 1.00

Flag question

When administering rTMS, which of the following frequency of stimulation is considered to have an excitatory effect on neuronal cells?

Select one:

- 0.5-0.9 Hz
- 5-20 Hz
- 2-4 Hz
- 1-5 Hz
- 3-7 Hz

High-frequency stimulation (5–20 Hz) is considered excitatory and low-frequency stimulation (1–5 Hz) inhibitory.

The correct answer is: 5-20 Hz

Question **46**

Not answered

Marked out of 1.00

Flag question

Which of the following is an endogenous process used to regulate gene expression and to degrade foreign mRNA from phages?

Select one:

- Nuclease Transcription
- Histone modification
- RNA interference
- DNA interference
- Spontaneous mutations

RNAi is an endogenous process used to regulate gene expression and to degrade foreign mRNA from phages. A double-stranded RNA (dsRNA) is cleaved by an enzyme called dicer resulting in small duplex fragments, (~ 19-23 base pairs long), called small interfering RNA (siRNA). The siRNA-Dicer forms a RNA-induced silencing complex (RISC) which further degrades RNA strands that have a complementary nucleotide sequence to one of the siRNA

guide strands. We can use a dsRNA molecule homologous to the mRNA transcript to be silenced to achieve the intended RNAi. We can also deliver synthetic siRNAs to attract the RISC complex and bind the homologous mRNA transcript. Also, short hairpin RNA (shRNA) with complementary sequences at either end, can cause the molecule to fold onto itself, producing a hairpin shape.

The correct answer is: RNA interference

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